Winnower: A Fast Tool for Splitting Cellular Automaton Spacetimes into a Periodic Background and a Residual Mask

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Abstract

Cellular automata often form broad periodic domains together with interfaces, defects, and particle-like activity. Winnower is an open-source tool that splits a finite CA spacetime into two parts: a *background* — the best-fitting periodic pattern, possibly drifting sideways over time — and a *residual mask* marking every cell the background fails to explain. The tool scans a grid of candidate periodicities, fits each one in a single linear pass by majority vote, and picks between them with a complexity-aware score (Bernoulli normalized maximum likelihood, NML) so that bigger candidates do not win just by having more parameters. On planted ground truth the selector recovers the true period and drift essentially always at realistic noise levels, and on shuffled and random controls of a representative rule panel it never reports periodic structure. Across 1D, 2D, and 3D rules the resulting residual masks isolate particle-like activity — for example in Diamoeba, Maze with Mice, and S37/B11 — cheaply enough for whole-catalog surveys. Every number and figure in this paper regenerates from a deterministic public pipeline, and an in-browser demo re-runs the identical code on any rule in the supported families, at any seed and horizon, checking the output against the committed results column by column wherever a committed row exists.

Introduction

Cellular automata (CA) frequently produce spacetime patterns in which broad regular domains coexist with defects, interfaces, and particle-rich activity. For ALIFE work those domains matter because they are the canvas against which coherent structures become visible: a glider is only a glider relative to the background it moves through. A practical question follows immediately: given a finite spacetime, which global periodic organization best explains it, and what remains once that organization is removed?

Two established traditions address nearby questions. Computational mechanics and local causal-state methods recover local statistical structure and can identify domains, particles, and interactions in rich detail (Crutchfield and Hanson, 1997; Hanson and Crutchfield, 1992; Shalizi et al., 2006; Lizier et al., 2008; Rupe and Crutchfield, 2018; Rupe et al., 2019). Survey-oriented and compression-oriented work instead emphasizes broad rule catalogs and low-cost global summaries (Packard and Wolfram, 1985; Boccara and Roger, 1999; Zenil, 2010; Zenil et al., 2018). Winnower sits between the two. It does not replace local structure reconstruction; it is a cheap global *front end* that can be applied across hundreds of rules, handing back a compact background hypothesis plus a residual mask that shows where richer local analysis is likely to pay off. The need for this kind of fast triage has only grown as newer substrates — Lenia, neural cellular automata — generate large volumes of spacetime data (Chan, 2019; Mordvintsev et al., 2020).

What the tool does, in plain terms. Winnower takes a binary spacetime U (a $T \times$ space array) and a grid of *candidates*. A candidate is a pair $(p, \mathbf{s})$: "the pattern repeats every p time steps, shifted sideways by $\mathbf{s}$ cells." For each candidate the tool groups cells that the candidate says must be equal (the *orbit classes*), takes a majority vote inside each group to build the best-fitting background $B^*(p, \mathbf{s})$, and records the *residual mask* $M = U \oplus B^*$ — the cells where the observation disagrees with its background. Candidates are then ranked by a score that adds a complexity penalty to the raw disagreement, and the winner is reported together with its margin over the runner-up. Each candidate costs one linear pass over the data, which is what makes whole-catalog surveys practical: scanning the default 130-candidate 1D grid on an 800×192 ($T \times$ space) ECA spacetime takes a few seconds single-threaded, and about half a second at T=200 — per-run timings ship in the released benchmark CSVs — while the same pipeline handles 2D and 3D volumes unchanged.

This paper contributes, in order of importance: (i) the tool itself, open source with a deterministic experiment pipeline and an in-browser reproduction of every reported number; (ii) evidence that the selector is accurate — it recovers planted ground truth, reports nothing on shuffled and random controls, and its choices match known CA backgrounds where those exist; (iii) surveys across 1D, 2D, and 3D rule catalogs showing that the background-plus-residual split isolates particle-like structure at survey scale; and (iv) two supporting mathematical properties, stated informally here with formal versions in the repository, that explain *why* the complexity penalty is necessary and why selections stabilize as the observation window grows.

The Winnower Pipeline

Vocabulary. Throughout: the *horizon* T is the number of observed time steps; a *candidate* $(p, \mathbf{s})$ is a hypothesized temporal period p and spatial shift vector $\mathbf{s}$; the *background* B^* is the best-fitting field that repeats with that periodicity; the *residual mask* M marks cells where U and B^* disagree (the code and released data tables call this the *defect* mask, e.g. column `defect_rate`); the *margin* is the score difference in bits between the winning candidate and the runner-up.

Candidates and orbit classes. For a binary spacetime $U \in \{0,1\}^{T \times D_1 \times \cdots \times D_n}$, a candidate $(p, \mathbf{s})$ with $p \geq 1$ and $\mathbf{s} \in \mathbb{Z}^n$ describes the set of fields B satisfying

$$B[t+p,\, \mathbf{x} + \mathbf{s} \bmod \mathbf{D}] = B[t, \mathbf{x}]$$

(a field satisfying this relation is called *relative-periodic*). Iterating the relation partitions the cells of U into orbit classes $\{O_j\}_{j=1}^{k}$ with $k = p \prod_i D_i$: two cells are in the same class exactly when any B obeying the candidate must give them the same value. Fitting a candidate therefore means choosing one bit per orbit class.

Proposition 1 (Majority vote is optimal)**.** *For a fixed candidate $(p, \mathbf{s})$, the background that minimizes the number of residual cells is obtained by taking a majority vote within each orbit class. Writing $n_j^{(1)}$ and $n_j^{(0)}$ for the ones and zeros in class O_j, the minimum residual count is $\sum_j \min(n_j^{(0)}, n_j^{(1)})$.*

This is the computational core of the tool: the disagreement objective decomposes over orbit classes, each class is an independent one-bit choice, and the whole fit is one linear pass over the data. (Proof in the repository's theory notes.)

Scoring candidates. Raw disagreement cannot be used to pick *between* candidates, because larger candidates fit better for free (next paragraph). Instead each orbit class is treated as a Bernoulli estimation problem with empirical rate $\hat{\theta}_j = n_j^{(1)}/n_j$, $n_j = |O_j|$, and a candidate is scored by

$$\mathrm{NML}(p, \mathbf{s}) \;=\; \underbrace{\sum_j n_j H_b(\hat{\theta}_j)}_{\text{fit (NLL)}} \;+\; \underbrace{\sum_j C(n_j)}_{\text{complexity}},$$

where H_b is binary entropy and $C(n)$ is the Bernoulli NML (Shtarkov) parametric complexity of a class with n observations (Shtarkov, 1987; Grünwald, 2007; Grünwald and Roos, 2019). The released implementation computes $C(n)$ by the exact Shtarkov sum for classes with $n \leq 200$ and by the asymptotic expansion $C(n) \approx \frac{1}{2}\log_2(\pi n/2)$ above that cutoff (the `hybrid` mode); this is the score used in every experiment below, and at survey horizons most orbit classes fall in the exact regime. The default search grids are: 1D, periods 1–10 and shifts $-6..6$ (130 candidates); 2D, periods 1–8 and shifts ± 2 per axis; 3D, periods 1–4 and shifts ± 2 per axis. The selector is *period-first*: it takes the best-scoring shift at each period, ranks periods by that best score, and reports the winning period, its shift, and the *margin* — the score gap in bits to the runner-up period. All ties are broken deterministically (smaller period, then smaller shift; majority votes break toward ones), and every observed tie and near-tie in our experiments is enumerated in a released stress report.

Why the penalty is necessary (informal). Call $(p_2, \mathbf{s}_2)$ a *velocity-matched multiple* of $(p_1, \mathbf{s}_1)$ when $p_2 = mp_1$ and $\mathbf{s}_2 = m\mathbf{s}_1 \bmod \mathbf{D}$ — the same drift velocity expressed with a longer period. A velocity-matched multiple splits every orbit class of the smaller candidate into finer pieces, so its optimal residual count and its NLL can only stay equal or improve. Unpenalized selection therefore drifts toward the largest available candidate whether or not extra periodic structure exists. This is not hypothetical: in the ablation of Figure 4, plain NLL picks the largest scanned period for *every* rule tested. The formal statement and proof are in the repository's theory notes.

Why selections stabilize (informal). For a fixed finite candidate grid, the fit term grows linearly with the number of observed cells while the complexity term grows only logarithmically with the horizon. So if one candidate has a strictly better asymptotic per-cell fit rate, it eventually beats every other candidate, and the selection locks in with a growing margin. In practice this is visible as "margin grows after locking" in Figure 3. For an exactly periodic background and its velocity-matched multiples — candidates with identical asymptotic rates — the complexity term provably breaks the tie toward the smaller period. (Equal-rate candidates at the *same* period but different shifts have identical complexity, so no tie-break exists there; the released stress report tracks every such tie we observed.) Formal statements and proof notes for these claims ship with the repository, together with Lean 4 artifacts at documented levels of completeness (the majority-vote optimality proposition and the stabilization core are fully machine-checked in public CI; the rest are clearly-labeled drafts); we deliberately keep the formal material out of the main text because the contribution this paper argues for is the tool, not the theory.

Results

Accuracy: Planted Ground Truth and Null Controls

A detection tool needs two calibration answers before any survey result is interpretable: does it find periodicity that is really there, and does it stay silent when none exists?

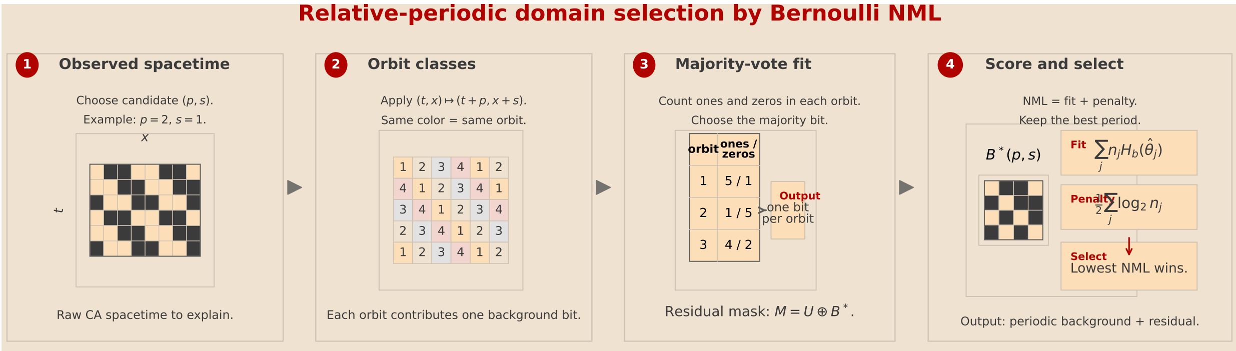

Figure 1: The Winnower pipeline on a small 1D example. (1) The observed spacetime U (time running downward). (2) A candidate $(p, \mathbf{s})$ groups cells into orbit classes: cells with the same number must be equal if the candidate is right. (3) A majority vote inside each class produces the background $B^*(p, \mathbf{s})$; cells that lost their vote form the residual mask $M = U \oplus B^*$. (4) Each candidate is scored by fit (Bernoulli negative log-likelihood, NLL) plus a complexity penalty; the candidate with the lowest total (Bernoulli NML) wins.

Table 1: Ground-truth recovery on planted relative-periodic spacetimes with bit-flip noise ε (642 runs; shipped selector, default search grids). Joint accuracy requires recovering both the period and the shift exactly.

Setting	ε	Runs	Joint acc.	Mean margin (bits)
1D (192, T=200)	0–0.2	480	1.00	14,889–4,021
2D (32^2, T=60)	0–0.1	108	1.00	21,793–11,699
3D (12^3, T=20)	0–0.1	54	1.00	9,883–5,485

Planted ground truth. We generate spacetimes that are exactly relative-periodic by construction — a random base tile extended by the candidate relation — with known $(p, \mathbf{s})$, then corrupt them with i.i.d. bit flips at rate ε and ask the shipped selector (default search grids, no tuning) to recover the truth. Table 1 summarizes the result: the selector recovers the planted period *and* shift in all 642 runs — every dimension, every planted $(p, \mathbf{s})$ in the scanned grid, and every tested noise level (up to $\varepsilon = 0.2$ in 1D, $\varepsilon = 0.1$ in 2D and 3D). Noise does not cause errors at these levels; it shows up as a shrinking decision margin (in 1D, from a mean of 14,889 bits at ε=0 down to 4,021 bits at ε=0.2), which is exactly the behavior a calibrated confidence signal should have. Recovery must eventually fail as $\varepsilon \to 0.5$; at the levels tested, majority votes over orbit classes of hundreds of cells remain correct with high probability.

Null controls. For every rule in the representative panel we also ran the selector on three destructive controls: time-shuffled frames, space-shuffled cells, and density-matched i.i.d. Bernoulli noise. Every single control run — across 1D, 2D, and 3D — selects period 1, i.e. no periodic organization, while the original spacetimes in the same panel select periods up to 4. In 2D the separation is stark: originals leave a mean residual rate of 1.4%, versus 38% for space-shuffled and Bernoulli controls of the same density. The tool does not hallucinate structure in noise.

Representative Decompositions

Figure 2 shows the main empirical use case: across dimensions, the selector removes the broad regular background and leaves behind exactly the defects and interfaces the background cannot explain. In the 2D named rules most relevant to ALIFE interpretation, Diamoeba (B35678/S5678) and Maze with Mice (B37/S12345) produce low-period backgrounds with concentrated residual masks, while S37/B11 retains a visibly organized residual population after background removal. (Named LifeWiki rules use B/S rulestring notation; rules such as S37/B11 come from a separate range-rule scan and are named S⟨survive range⟩/B⟨birth range⟩.) The Diamoeba panel shows a run on the period-2 side of a near-boundary selection; across seeds its modal period is 1 (Section tables and the released per-seed data make such boundary cases explicit).

In 1D the tool's choices can be checked against fifty years of ECA analysis. Rule 54, whose regular background and particle interactions are the canonical example in the computational-mechanics literature (Hanson and Crutchfield, 1992; Crutchfield and Hanson, 1997), locks to a period-4, shift-0 background in 10 of 10 seeds by $T = 200$, and the resulting residual mask collapses onto the familiar particle world-lines. Rule 110 is genuinely more ambiguous and the tool reports that ambiguity rather than hiding it: at $T = 800$, five of ten seeds select a period-7, zero-shift candidate — recovering the temporal period of the Rule 110 ether (Cook, 2004) — while four select a period-4, shift -2 alternative. Rule 30, as expected for an effectively aperiodic rule, stays at period 1 with near-arbitrary shift. A direct quantitative comparison against local causal-state domain filters on shared examples is natural future work.

Why the Penalty Matters

Figure 4 is the key practical argument for the complexity penalty. This ablation uses a single-seed run over the 105

Table 2: Life-like survey: modal selected period across five seeds, all 106 named LifeWiki rules.

Modal period	T=100	T=200	T=400	T=800
$p = 1$	98	99	96	93
$p = 2$	7	5	8	11
$p = 4$	0	1	1	1
$p = 8$	1	1	1	1

Life-like rules that neither die out nor freeze (the survey's triviality filter), so its counts differ slightly from Table 2's five-seed modal counts. At $T = 100$, plain NLL selects the largest tested period for every single rule — exactly the failure mode predicted above — and raw residual minimization shows the same upward pull (81 of 105 rules at period ≥ 4). Bernoulli NML instead selects period 1 for 97 rules, period 2 for 7, and period 8 for 1. The penalty is doing real work: it prevents the selector from confusing representational capacity with periodic structure.

Cross-Dimensional Behavior and 2D Surveys

We surveyed all 106 named Life-like rules from the LifeWiki on a 64×64 torus with periods 1–8 and shifts ±2 per axis, running five seeds per rule at horizons $T \in \{100, 200, 400, 800\}$ and reporting each rule's modal (across-seed majority) period. Table 2 shows the distribution: the large majority of rules stay at period 1 at every horizon, and the set of rules with periodic organization grows slowly and non-monotonically with observation time — from 8 rules at $T = 100$ to 13 at $T = 800$, dipping to 7 at $T = 200$ as a transient-period rule drops out. Eight rules change modal period across horizons: B017/S01 climbs from period 2 to period 4; Serviettes (B234/S) drops from 2 to 1 as an initial transient dies out; and six rules (Bacteria, Invertamaze, LowLife, Maze with Mice, Mazectric with Mice, Snakeskin) reveal period-2 organization only at longer horizons. Fredkin (B1357/S02468) is the only named rule that reaches period 8. Some rules sit near the decision boundary across seeds — Diamoeba, for instance, stays at modal period 1 while one of five seeds moves to period 2 at $T \geq 400$ — and the released per-seed tables expose every such case.

The same stabilization picture appears across dimensions with different transients. In 2D persistent-residual rules, longer horizons can reveal additional periodic structure before the selector locks. In 3D diamoeba3d, an initial period-6 selection is a small-sample artifact and period 1 takes over by $T = 40$ with growing margin. Selections change exactly when a longer observation window uncovers fit improvements strong enough to beat the logarithmic complexity cost.

Structured Residuals as the ALIFE Payoff

The value of the split is not the period number itself. The background acts as a domain hypothesis, and the residual mask highlights where particles, interfaces, and other coherent departures live. In Diamoeba and Maze with Mice

Table 3: Compact cross-dimensional summary of selected backgrounds.

Setting	Main observation
1D ECA 30/54/110	Rule 54 locks to period 4 (10/10 seeds); Rule 110 splits between period 7 (5/10) and period 4 with shift -2 (4/10) at $T = 800$; Rule 30 stays at period 1.
2D Life-like survey	At $T = 800$, 93 of 106 named rules keep modal period 1; 13 show periodic organization (11 at p=2, one each at p=4 and p=8).
2D focal persistent-residual rules	By T=800, S24/B11 and S37/B11 move to period 2 and S11/B37 to period 4; all retain organized residual structure after subtraction.
3D diamoeba3d	A short small-sample transient favors period 6; the selector stabilizes to period 1 by $T = 40$.

the residual is sparse and clearly localized. The focal range rules (S24/B11, S11/B37, S37/B11) were themselves *found by* an earlier screening pass of this decomposition over 773 totalistic range rules — the tool surfacing its own best examples is precisely the intended survey workflow, and we state it plainly so the reader does not mistake them for an independent sample. What the follow-up runs then establish is robustness: S37/B11's residual population persists at 400 steps, across 10 seeds, and at grid sizes up to 192×192, rather than dissolving as a transient. Whether a residual mask is "organized" is judged visually in Figure 2; quantifying residual structure (and adding a mask-structure null control) is future work. Even so, the decomposition already works as survey-scale triage — it points at the rules, horizons, and spatial regions that merit the much heavier cost of local causal-state or particle-level analysis.

Reproducibility and Artifact

Everything reported here regenerates from a public, deterministic pipeline:

- **Code and data:** `https://github.com/zitterbewegung/Winnower`. `make data` reruns the full experiment suite from base seed 11 (null controls, seed stability, ground truth, surveys, stress tests); `make figures` regenerates every figure; `make test` runs the 364-test suite. Each experiment writes a manifest recording its exact parameters.
- **Reviewer site:** `https://zitterbewegung.github.io/Winnower/` presents the claim ledger, all figures with their generating scripts, and sortable result tables built from the same CSVs as this paper.
- **Run it in your browser:** `https://zitterbewegung.github.io/Winnower/reproduce.html` runs the *identical* pipeline code in WebAssembly Python on any ECA rule, any 2D B/S rulestring, or any 3D birth/survive rule, and compares the outcome against the committed result row for the

same rule, seed, and horizon, column by column at 10^{-9} relative tolerance. The browser code path is verified bit-compatible with the committed results by `scripts/verify_webdemo_bootstrap.py`.

- **Formal notes:** the repository's `docs/` contains a claim ledger auditing every mathematical statement in the project's long-form theory notes, with an explicit map to this paper's statements. `proofs/` contains Lean 4 artifacts generated with the Aristotle prover at documented levels of completeness: placeholder-free, in-build, CI-checked proofs of Proposition 1 and of the stabilization property's algebraic core (against a pinned Mathlib toolchain), plus clearly-labeled work-in-progress drafts for the remaining statements; `proofs/README.md` records the per-file status. We make no claim in this paper that depends on the Lean artifacts.

Discussion

Winnower is deliberately modest. It is complementary to computational mechanics rather than a replacement: the output is a compact global background and a residual mask, not an account of interaction structure. What it buys in exchange is scale — the same orbit-class computation runs unchanged across many rules and across 1D, 2D, and 3D.

Its limitations are equally concrete. The selector returns the best-compressing global explanation, which need not coincide with a physicist's preferred background when the residual itself carries periodic structure. Period-1 shift ties make the reported shift arbitrary for effectively aperiodic rules. The scanned grid bounds what can be found: periodicity outside the grid is invisible, as the candidate-range robustness runs in the repository quantify. And the residual mask says where structure is, not how it interacts. Natural next steps are direct comparisons with local causal-state filters on shared examples, component tracking within residual masks, and application to continuous substrates such as Lenia after thresholding.

Conclusion

Winnower turns a simple observation — relative-periodic fitting reduces to independent majority votes — into a practical survey instrument: linear-time background fitting, a complexity-aware selector that resists the period inflation raw fit criteria demonstrably suffer (with a prose proof, audited in the repository's claim ledger, that it resists it whenever the background is exactly periodic), calibration on planted truth and null controls, and a residual mask that isolates particle-like structure across 1D, 2D, and 3D. The tool, the data, the proofs, and an in-browser reproduction of every reported number are public. For ALIFE the payoff is practical: screen many rules cheaply, find where coherent structure lives, and spend expensive analyses only where this cheap one says they will pay off.

Acknowledgements

Language-model assistants (OpenAI ChatGPT and Anthropic Claude) were used for drafting text, code, and proof sketches; the Aristotle prover generated the Lean artifacts. All experimental numbers come from the deterministic pipeline in the public repository and are independently validated by its test suite and by the in-browser reproduction. The prose mathematical claims were audited by hand in the repository's claim ledger; the per-file status of the Lean artifacts (what is machine-checked and what remains a draft) is documented in `proofs/README.md`.

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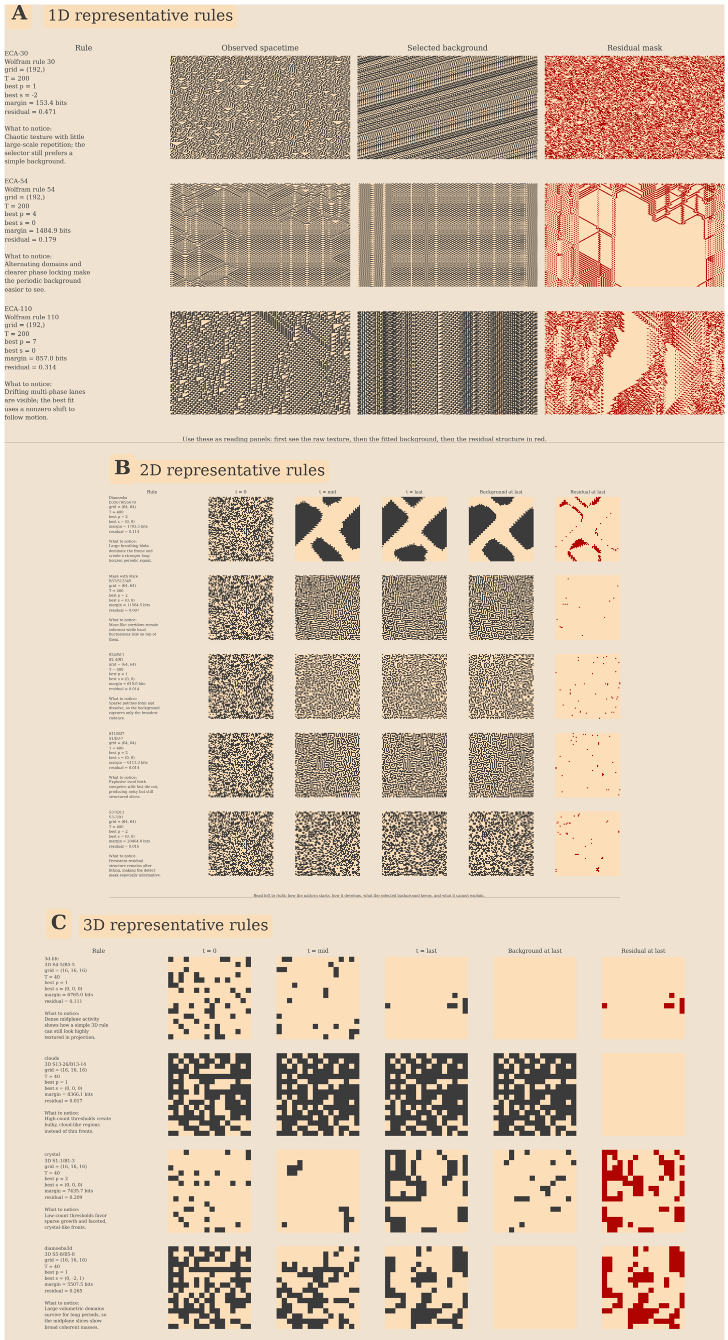

Figure 2: Representative decompositions across 1D, 2D, and 3D: observed spacetime U, selected background B^*, and residual mask M for each rule; in 1D panels time runs downward and space is horizontal. Each panel's sidebar reports the winning candidate (best p, best $\mathbf{s}$), its margin over the runner-up in bits, and the residual rate (the fraction of cells in M). In the focal 2D cases the residual stays sparse but visibly organized rather than collapsing to featureless noise.

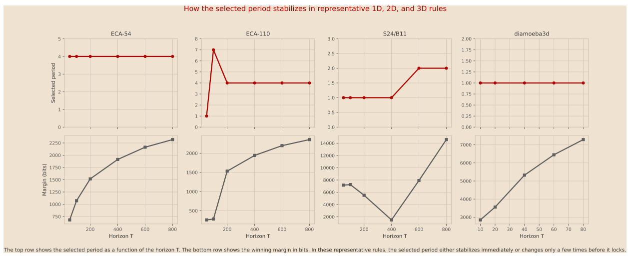

Figure 3: Selected period versus horizon T (number of observed time steps) for representative rules. The selection typically locks early, and the NML margin between winner and runner-up (in bits) grows after locking.

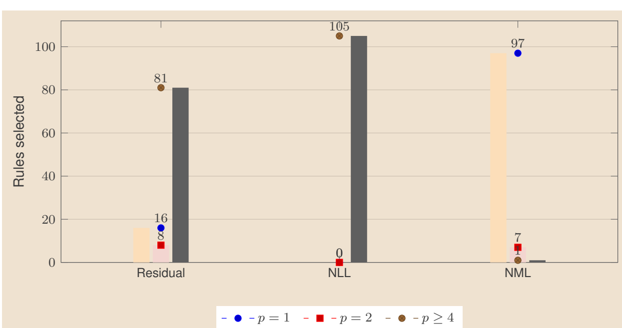

Figure 4: Selector ablation on the 105 non-degenerate Life-like rules at $T = 100$ (single seed). Bars count rules by the period each criterion selects. Raw residual minimization and plain Bernoulli NLL are pulled toward the largest scanned period; Bernoulli NML concentrates selections at low period.